Problem Set 1

Econometrics

Code: https://github.com/ae4292/Econometrics_PSets

Problem 1

(a) $f_{YA}(y,a) = \frac{1}{4}$. With $Y = A\sqrt{L}$, the Jacobian of the inverse transformation $g^{-1}(A,L) = (Y,A)$ is

$$J = \begin{pmatrix} \frac{\partial A}{\partial A} & \frac{\partial A}{\partial L} \\ \frac{\partial Y}{\partial A} & \frac{\partial Y}{\partial L} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ \sqrt{L} & \frac{A}{2\sqrt{L}} \end{pmatrix}, \qquad |J| = \frac{A}{2\sqrt{L}}.$$

Since $0 \le Y \le 2$ means $0 \le A\sqrt{L} \le 2$,

$$f_{AL}(A,L) = f_{YA}(g^{-1}(A,L))\,|J| = \frac{A}{8\sqrt{L}}, \qquad 0 \le A \le 2, \quad 0 \le L \le \left(\frac{2}{A}\right)^2$$

(b) The causal effect is $\partial Y/\partial L = A/(2\sqrt{L})$, which is $A/2$ at $L = 1$. The ACE averages this conditional on L. At $L = 1$, $f_L(1) = \int_0^2 \frac{A}{8}\,dA = \frac{1}{4}$, so $f_{A|L}(A \mid 1) = \frac{A/8}{1/4} = \frac{A}{2}$ and $\mathbb{E}(A \mid L = 1) = \int_0^2 A \cdot \frac{A}{2}\,dA = \frac{4}{3}$. Therefore

$$\mathrm{ACE}(L = 1) = \tfrac{1}{2}\,\mathbb{E}(A \mid L = 1) = \frac{2}{3}$$

(c) $A \le 2$ and $A \le 2/\sqrt{L}$ must both hold, so integrating out A over $[0, \min(2, 2/\sqrt{L})]$ gives

$$f_L(L) = \begin{cases} \dfrac{1}{4\sqrt{L}}, & L \le 1, \\ \dfrac{1}{4L^{3/2}}, & L > 1, \end{cases}$$

$$\mathbb{E}(A \mid L) = \frac{1}{f_L(L)} \int_0^{\min(2,2/\sqrt{L})} \frac{A^2}{8\sqrt{L}}\,dA = \begin{cases} \dfrac{4}{3}, & L \le 1, \\ \dfrac{4}{3\sqrt{L}}, & L > 1. \end{cases}$$

Since $\mathbb{E}(Y \mid L) = \sqrt{L}\,\mathbb{E}(A \mid L)$,

$$\mathbb{E}(Y \mid L) = \begin{cases} \frac{4}{3}\sqrt{L}, & L \le 1, \\ \frac{4}{3}, & L > 1. \end{cases}$$

so $\frac{d}{dL}\mathbb{E}(Y \mid L)$ is $\frac{2}{3}$ at $L = 1$ from the left and 0 from the right.

Comparison with (b). For $L \le 1$ the slope of the CEF equals the ACE, because $f_{A|L} = A/2$

does not depend on L. For $L > 1$ the support of $A \mid L$ is $[0, 2/\sqrt{L}]$, which depends on L, so the slope of the CEF is not the ACE.

(d) $b = \mathrm{Cov}(Y, L)/\mathrm{Var}(L)$ requires L to have finite variance, but

$$\mathbb{E}(L) = \mathbb{E}(Y^2)\,\mathbb{E}(A^{-2}) = \frac{4}{3}\int_0^2 \frac{1}{2}a^{-2}\,da = \infty, \qquad \mathbb{E}(L^2) = \mathbb{E}(Y^4)\mathbb{E}(A^{-4}) = \infty.$$

So $\mathbb{E}(Y - a - bL)^2 = \infty$ for every $b \neq 0$, and the minimum is at $b = 0$:

$$b = 0, \qquad a = \mathbb{E}(Y) = 1$$

Problem 2

(a)

$$\mathbb{E}(y \mid x) = \mathbb{E}\big(f(x) + x\epsilon \mid x\big) = f(x) + x\,\mathbb{E}(\epsilon \mid x) = f(x)$$

as $\mathbb{E}(\epsilon) = 0$ and $x \perp\!\!\!\perp \epsilon$.

(b)

$$\mathrm{Var}(y \mid x) = \mathbb{E}(y^2 \mid x) - \big(\mathbb{E}(y \mid x)\big)^2,$$

$$\begin{aligned}\mathbb{E}(y^2 \mid x) &= \mathbb{E}\big(f(x)^2 + 2f(x)x\epsilon + (x\epsilon)^2 \mid x\big) \\ &= f(x)^2 + 2f(x)x\,\mathbb{E}(\epsilon) + x^2\mathbb{E}(\epsilon^2) = f(x)^2 + x^2 \cdot 1,\end{aligned}$$

$$\mathrm{Var}(y \mid x) = f(x)^2 + x^2 - f(x)^2 = x^2.$$

$$\mathrm{MSE} = \mathbb{E}\big(\,\mathrm{Var}(y \mid x) + (\mathrm{Bias})^2\big), \quad \text{bias is zero in this case,}$$

$$\mathrm{MSE} = \mathbb{E}\big(\,\mathrm{Var}(y \mid x)\big) = \mathbb{E}(x^2) = 1, \qquad x \sim N(0,1).$$

$$\mathrm{Var}(y \mid x) = x^2, \qquad \mathrm{MSE}_{\mathrm{CEF}} = 1$$

(c) Best linear predictor $\alpha + \beta x$:

$$\alpha = \mathbb{E}(y) - \beta\,\mathbb{E}(x) = \mathbb{E}(y) - \beta \cdot 0 = \mathbb{E}(f(x)), \qquad \beta = \frac{\mathrm{Cov}(x,y)}{\mathrm{Var}(x)} = \frac{\mathrm{Cov}(x,y)}{1},$$

$$\mathrm{Cov}(x,y) = \mathbb{E}(xy) - \mathbb{E}(x)\mathbb{E}(y) = \mathbb{E}(xy) - 0,$$

$$\mathbb{E}(xy) = \mathbb{E}\big(xf(x) + x^2\epsilon\big) = \mathbb{E}\big(xf(x)\big), \qquad xf(x) = \begin{cases} x^2 & x \geq 0 \\ 0 & x < 0 \end{cases}, \quad P(x \geq 0) = \tfrac{1}{2},$$

$$\mathbb{E}\big(xf(x)\big) = \mathbb{E}\big(x^2 \mid x \geq 0\big)P(x \geq 0) = 1 \cdot \tfrac{1}{2} = \tfrac{1}{2} \;\Rightarrow\; \mathrm{Cov}(x,y) = \tfrac{1}{2}.$$

$$\mathbb{E}\big(f(x)\big) = \int_0^\infty x \cdot \frac{1}{\sqrt{2\pi}}e^{-x^2/2}\,dx = -\frac{1}{\sqrt{2\pi}}e^{-x^2/2}\Big|_0^\infty = 0 + \frac{1}{\sqrt{2\pi}} \;\Rightarrow\; \alpha = \frac{1}{\sqrt{2\pi}}.$$

$$\alpha = \frac{1}{\sqrt{2\pi}}, \qquad \beta = \frac{1}{2}$$

$$\text{MSE} = \mathbb{E}\Big(y - \frac{1}{\sqrt{2\pi}} - \tfrac{1}{2}x\Big)^2 = \mathbb{E}(y^2) + \frac{1}{2\pi} + \frac{1}{4}\mathbb{E}(x^2) - \frac{2}{\sqrt{2\pi}}\mathbb{E}(y) - \mathbb{E}(yx) + \frac{\mathbb{E}(x)}{\sqrt{2\pi}},$$

with $\mathbb{E}(y^2) = \mathbb{E}\big((f(x) + x\epsilon)^2\big) = \mathbb{E}\big(f(x)^2\big) + 1 = \frac{1}{2}\mathbb{E}(x^2) + 1 = \frac{1}{2} + 1$, so

$$\text{MSE} = \tfrac{1}{2} + 1 + \frac{1}{2\pi} + \frac{1}{4} - \frac{2}{\sqrt{2\pi}} \cdot \frac{1}{\sqrt{2\pi}} - \frac{1}{2} = \frac{5}{4} - \frac{1}{2\pi} = 1.091.$$

$$\text{MSE}_{\text{BLP}} = \frac{5}{4} - \frac{1}{2\pi} \approx 1.091$$

(d) For a step function $g(x)$ (not minimisable with respect to $g(x)$ in the first term),

$$\begin{aligned}\text{MSE} &= \mathbb{E}(y - g(x))^2 \\ &= \underbrace{\mathbb{E}\big(y - \mathbb{E}(y \mid x)\big)^2}_{=1} + \mathbb{E}\big(\mathbb{E}(y \mid x) - g(x)\big)^2,\end{aligned}$$

$$\begin{aligned}\mathbb{E}\big(\mathbb{E}(y \mid x) - g(x)\big)^2 &= \mathbb{E}\big(f(x) - g(x)\big)^2 \\ &= \int_{-\infty}^{x_0} \big(f(x) - a\big)^2 \phi(x)\, dx + \int_{x_0}^{\infty} \big(f(x) - b\big)^2 \phi(x)\, dx\end{aligned}$$

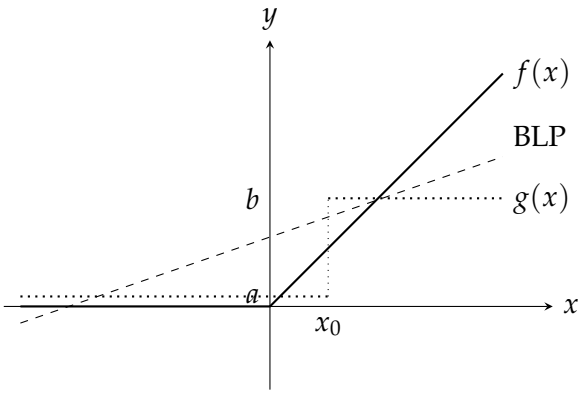
where ϕ is the standard normal density. Minimising numerically (code: ps1_cart.py) gives

$$x_0 = 0.7049, \qquad a = 0.1155, \qquad b = 1.2942,$$

$$\text{MSE}_{\text{CART}} = 1 + \mathbb{E}\big(f(x) - g(x)\big)^2 = 1 + 0.0871 = 1.0871.$$

$$\text{MSE}_{\text{CART}} = 1.0871$$

The MSE of the BLP is higher than that of the CART.



Problem 3

(a) With one observation,

$$\hat{\beta}_{OLS} = \Big(\sum x_i x_i\Big)^{-1} \sum x_i y_i = \frac{y_i}{x_i}.$$

$$\mathbb{E}(\hat{\beta}_{OLS}) = \mathbb{E}\left(\frac{y}{x}\right) = \mathbb{E}\left(\frac{\beta x + \varepsilon}{x}\right) = \beta \cdot 1 + \mathbb{E}\left(\frac{\varepsilon}{x}\right) = \beta,$$

since $\mathbb{E}(\varepsilon \mid x) = 0$.

$$\mathrm{Var}(\hat{\beta}_{OLS}) = \mathrm{Var}\left(\frac{y}{x}\right) = \mathrm{Var}\left(\frac{\beta x + \varepsilon}{x}\right) = \mathrm{Var}(\beta) + \mathrm{Var}\left(\frac{\varepsilon}{x}\right) = 0 + \mathrm{Var}\left(\frac{\varepsilon}{x}\right),$$

$$\mathrm{Var}\left(\frac{\varepsilon}{x}\right) = \mathbb{E}\left(\frac{1}{x^2}\right)\mathbb{E}(\varepsilon^2) - \mathbb{E}\left(\frac{1}{x}\right)^2 \mathbb{E}(\varepsilon)^2 = \mathbb{E}\left(\frac{1}{x^2}\right) \cdot 1 - 0 = \frac{25 + \frac{25}{49}}{2} = \frac{625}{49}.$$

> $\mathbb{E}(\hat{\beta}) = \beta, \qquad \mathrm{Var}(\hat{\beta}) = \dfrac{625}{49}$

(b)

$$\mathbb{E}(\tilde{\beta}) = \mathbb{E}(xy) = \mathbb{E}(x\beta x + x\varepsilon) = \mathbb{E}(x^2\beta) = \beta\,\mathbb{E}(x^2), \qquad \mathbb{E}(x^2) = \tfrac{1}{2}\left(\tfrac{1}{25} + \tfrac{49}{25}\right) = \tfrac{1}{2} \cdot 2 = 1,$$

so $\mathbb{E}(\tilde{\beta}) = \beta$: $\tilde{\beta}$ is unconditionally unbiased for β.

$$\mathrm{Var}(\tilde{\beta}) = \mathrm{Var}(xy) = \mathrm{Var}(x\beta x + x\varepsilon) = \beta^2\,\mathrm{Var}(x^2) + \mathrm{Var}(x\varepsilon),$$

$$\mathrm{Var}(x^2) = \mathbb{E}(x^2) - \mathbb{E}(x)^2 = 1 - \tfrac{4}{5} = \tfrac{1}{5},$$

$$\mathrm{Var}(x\varepsilon) = \mathbb{E}(x^2\varepsilon^2) - \mathbb{E}(x)^2\mathbb{E}(\varepsilon)^2 = 1 \cdot 1 - 0 = 1,$$

> $\mathrm{Var}(\tilde{\beta}) = \dfrac{\beta^2}{5} + 1$

(c)

$$\mathrm{Var}(\hat{\beta}_{OLS}) = \frac{625}{49}, \qquad \mathrm{Var}(\tilde{\beta}) = \frac{0}{5} + 1 = 1 \quad (\beta = 0).$$

$\mathrm{Var}(\tilde{\beta}) < \mathrm{Var}(\hat{\beta}_{OLS})$, which are unconditional variances. Gauss–Markov says $\hat{\beta}_{OLS}$ is the best conditional estimator of β that minimises the variance. Thus it does not contradict, since these are unconditional estimates.

Problem 4

$$\begin{aligned}\mathrm{Var}(\tilde{\beta} \mid X) &= \mathrm{Var}\left(\frac{\hat{\beta}_1 + \cdots + \hat{\beta}_k}{k} \,\middle|\, X\right) \\ &= \frac{1}{k^2}\,\mathrm{Var}\left(\hat{\beta}_1 + \cdots + \hat{\beta}_k \mid X\right).\end{aligned}$$

Since the $\hat{\beta}_j$ are independent,

$$\frac{1}{k^2}\,\mathrm{Var}(\hat{\beta}_1 + \cdots + \hat{\beta}_k \mid X) = \frac{1}{k^2}\Big(\mathrm{Var}(\hat{\beta}_1 \mid X) + \cdots + \mathrm{Var}(\hat{\beta}_k \mid X)\Big).$$

Since they should have the same distribution,

$$\begin{aligned}\frac{1}{k^2}\Big(\operatorname{Var}(\hat{\beta}_1 \mid X) + \cdots + \operatorname{Var}(\hat{\beta}_k \mid X)\Big) &= \frac{1}{k}\operatorname{Var}(\hat{\beta}_1 \mid X) \\ &= \frac{1}{k}\hat{\sigma}^2 (X'X)^{-1} \\ &= \frac{1}{k} \cdot \frac{\hat{\epsilon}'\hat{\epsilon}}{\frac{n}{k} - 1} (X'X)^{-1} \\ &= \frac{1}{n-k}\,\hat{\epsilon}'\hat{\epsilon}\,(X'X)^{-1}.\end{aligned}$$

Therefore

$$\operatorname{Var}(\hat{\beta}_{OLS} \mid X) = \frac{1}{n-k}\hat{\epsilon}'\hat{\epsilon}\,(X'X)^{-1},$$

which is the same as $\operatorname{Var}(\tilde{\beta} \mid X)$ for bagging. This estimator is not better than the non-split sample, since they have the same estimator variance.

Problem 5

Identity

$$M_Z = I - Z(Z'Z)^{-1}Z', \qquad M_Z W = W - Z(Z'Z)^{-1}Z'W,$$

$$\begin{aligned}M_{M_Z W} &= I - M_Z W\big((M_Z W)' M_Z W\big)^{-1}(M_Z W)' \\ &= I - M_Z W\big(W' M_Z M_Z W\big)^{-1} W' M_Z \\ &= I - M_Z W\big(W' M_Z W\big)^{-1} W' M_Z,\end{aligned}$$

$$M_{M_Z W} M_Z = M_Z - M_Z W\big(W' M_Z W\big)^{-1} W' M_Z.$$

By Frisch–Waugh–Lovell,

$$\begin{aligned}M_{W,Z} &= M_Z + M_{M_Z W} - I \\ &= M_Z + I - M_Z W\big(W' M_Z W\big)^{-1} W' M_Z - I \\ &= M_Z - M_Z W\big(W' M_Z W\big)^{-1} W' M_Z.\end{aligned}$$

> $M_{W,Z} = M_{M_Z W}\, M_Z$

Implication for the 3-step procedure

1. $R_Y = M_Z Y, \quad R_X = M_Z X, \quad R_W = M_Z W.$
2. $\tilde{R}_Y = M_{M_Z W} M_Z Y, \quad \tilde{R}_X = M_{M_Z W} M_Z X.$
3. $\tilde{R}_Y = M_{W,Z} Y, \quad \tilde{R}_X = M_{W,Z} X$, so

$$M_{W,Z} Y = \beta \cdot M_{W,Z} X + \varepsilon, \qquad \varepsilon = M_{M_{W,Z} X} M_{W,Z} Y = M_{W,Z,X}\, Y.$$

$\hat{\beta}_1$ is orthogonal to W and Z, so it is essentially the effect of X on Y without W and Z.